

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-4095
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	5 Marks

Performance Testing

SECMS is a declarative Salesforce application whose infrastructure is managed by Salesforce, so performance was assessed through normal use of the Lightning interface rather than a load-testing tool. Values that were not measured are left for the group to record and are not filled in with assumed figures.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Salesforce Lightning UI (manual observation)
Type of Testing	Functional testing with basic responsiveness observation (NFR-4: quick search and dashboard refresh)
Target Module / API	Student and Enrollment validation rules, enrollment flows, approval process, reports and dashboard
Test Environment	Salesforce Developer Edition org (cloud hosted)
Test Date	____ (to be entered by the group)

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Create a student record (email validation rule applies)	1	____	Valid record saves; invalid email is rejected with a message
2	Create an enrollment and trigger automation (enrollment date, instructor assignment, follow-up task)	1	____	Enrollment date, instructor and task are populated
3	Submit enrollment for approval and approve it	1	____	Status changes to Approved when fees are paid
4	Open reports and the Student Management Dashboard	1	____	Reports and dashboard components load

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	Responsive UI operations	____	____	Record after execution
2	Response Time (Max)	Responsive UI operations	____	____	Record after execution
3	Throughput (Req/sec)	N/A	N/A	N/A	No API load test; declarative app

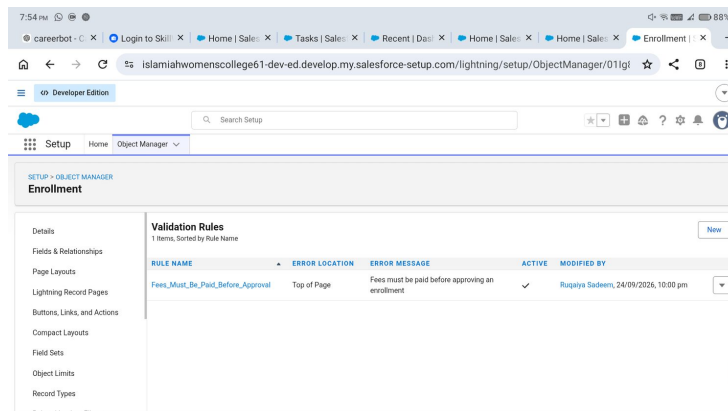
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
4	Error Rate	No critical errors	_____	_____	Record after execution
5	CPU Utilization	N/A	N/A	N/A	Platform-managed by Salesforce
6	Memory Utilization	N/A	N/A	N/A	Platform-managed by Salesforce

Step 4: Observations & Analysis

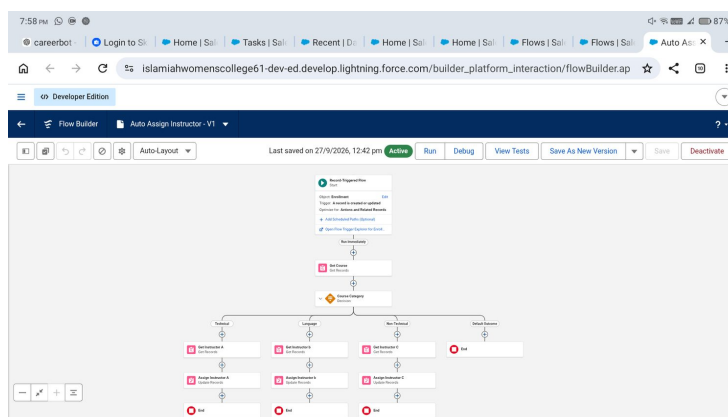
Core SECMS operations (record creation, validation, flows, approval, reports and dashboard) run on the Salesforce platform. The functional test cases in the project report cover these operations; TC-11 to TC-16 (validation and automation) are recorded as PASS. Server CPU and memory metrics are not exposed because the infrastructure is Salesforce-managed, and no dedicated load-testing tool was used.

Step 5: Screenshots / Evidence

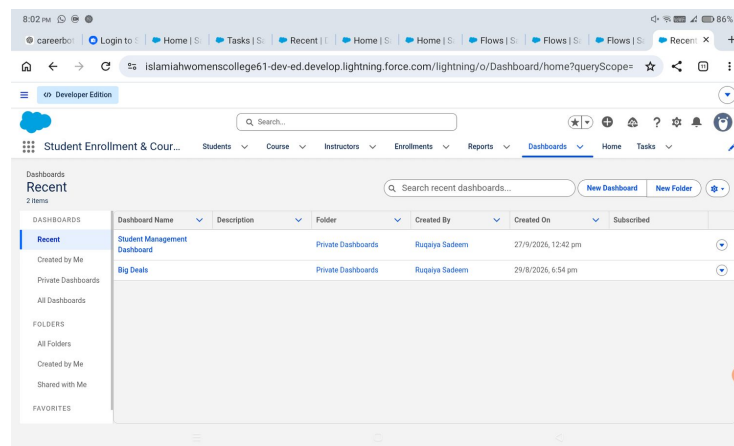
Salesforce evidence for the tested features is provided below and in the Screenshots folder.



Fees-paid validation rule on Enrollment



AutoAssign_Instructor flow



Student Management Dashboard